

Python 09

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Solving Linear Systems in Python

Numpy `linalg.solve`

`numpy.linalg.solve` solves a linear matrix equation, or system of linear scalar equations of the form:

$$\mathbf{Ax} = \mathbf{b} \quad (1)$$

where **A** is a square matrix and **b** is a vector.

Numpy `linalg.solve`

```
1 import numpy as np
2 A = np.array([[2, 3],
3               [3, 4]])
4
5 B = np.array([8, 11])
6
7 # Solve the system of equations
8 X = np.linalg.solve(A, B)
9
10 print("The solution is:")
11 print("x =", X[0])
12 print("y =", X[1])
```

Solving DEs and Eigenvalue-Problems in Python

Scipy `scipy.integrate.solve_ivp`

To solve initial value the `solve_ivp` function from the scipy can be used.

```
scipy.integrate.solve_ivp(fun, t_span, y0,  
method='RK45', t_eval=None, dense_output=False,  
events=None, vectorized=False, args=None)
```

Scipy `scipy.integrate.solve_ivp`

- `fun`: function of DE. `fun` takes 2 args
- `t_span`: tuple $(t_0, t_f) \rightarrow$ integration intervall
- `y0`: initial conditions (array of initial values)
- `method`: Integration-method (defaults to 'RK45')
- `t_eval`: Array of timesteps where sol is stored (opt.)
- ...

Scipy `scipy.integrate.solve_ivp`

```
1 from scipy.integrate import solve_ivp
2 # Define the ODE as a function
3 def dydt(t, y):
4     return -2 * y
5 # Initial condition
6 y0 = [1]
7 # Time points where we want the solution
8 t_span = (0, 5)
9 t_eval = np.linspace(0, 5, 100)
10 # Solve the ODE
11 solution = solve_ivp(dydt, t_span, y0,
12                       t_eval=t_eval)
13 t, y = solution.t, solution.y[0]
```

Numpy `linalg.eig`

To calculate the eigenvalues and eigenvectors for a given square matrix, the `numpy.linalg.eig` function can be used. `numpy.linalg.eig(a)`

- Input: a ... square matrix
- Output: (eigenvalues, eigenvectors) ... named tuple with eigenvalues and eigenvectors

Further numpy specialties/functions

Numpy @-operator

Starting with `python 3.5` `numpy` has a build-in operator for matrix multiplication.

Note: The `*`-operator does not matrix-multiply two 2D-arrays ($\rightarrow$ matrices)! It can be used to multiply element-wise.

To actually perform matrix multiplication the `@`-operator should be used.

Dot - products

The numpy.dot function calculates a dot product for two given arrays.

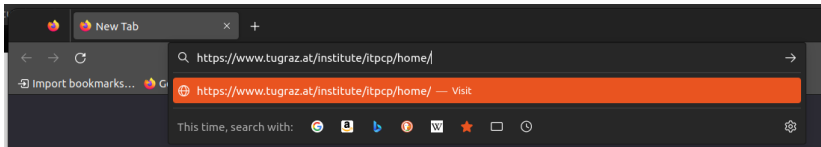
It may also be used to multiply two 0-D arrays (meaning numbers) or to multiply 2-D arrays (matrices). However, the `*`- and `@`-operators are preferred.

How do Computers communicate

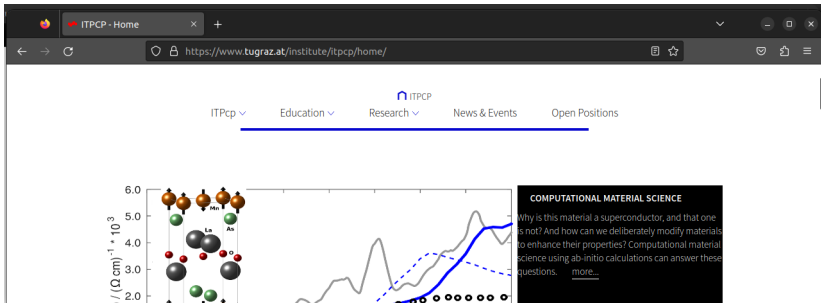
How do Computers communicate?

Computers can communicate with other remote computer by using:

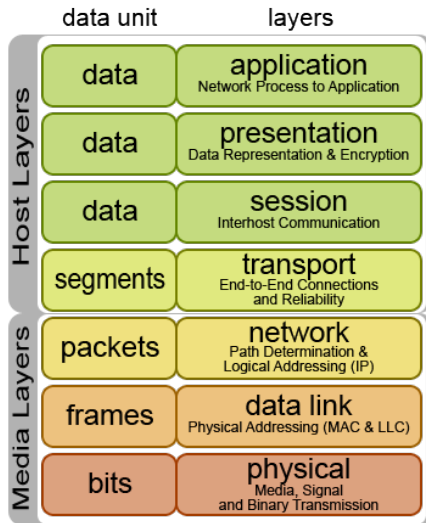
- **Ethernet**: Used to connect computers in a local network.
- **WiFi**: Wireless Fidelity. Used to connect computers in a local network.
- **TCP/UDP**: Transmission Control Protocol / User Datagram Protocol. Used to transfer data over the internet.
- **HTTP**: HyperText Transfer Protocol. Used to transfer web pages.



↓ How does this work? ↓



OSI Model



https://en.wikipedia.org/wiki/OSI_model

Let's look at a simplified example:

We want to access www.tugraz.at/institute/itpcp/

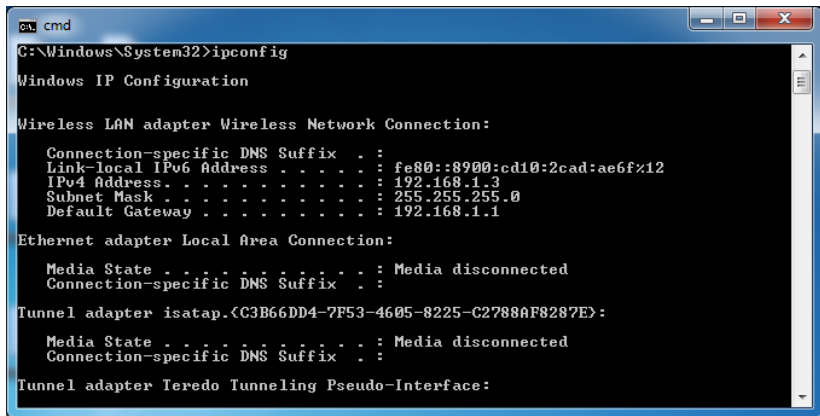
Our browser will send a **request** to the server at www.tugraz.at using the **HTTP** protocol.

But how does he know where to send the request?

Computers have addresses, just like houses. They are called **IP addresses**.¹

¹ actually it's more involved especially with IPv4 addresses → [IPv4-exhaustion](#)

IPv4 addresses



```
C:\ cmd
C:\Windows\System32>ipconfig

Windows IP Configuration

Wireless LAN adapter Wireless Network Connection:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::8900:cd10:2cad:ae6f%12
    IPv4 Address. . . . . : 192.168.1.3
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

Ethernet adapter Local Area Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter isatap.{C3B66DD4-7F53-4605-8225-C2788AF8287E}:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Tunnel adapter Teredo Tunneling Pseudo-Interface:
```

<https://www.flickr.com/photos/johnlsloan/5477351098>

How to get an IP address for a website?

Just ask another server for it → **DNS request** The

request will be send from your computers network card to your router.

The router will forward the request to your ISP (Internet Service Provider; A1, Magenta, etc.).

Maybe your ISP knows the IP address of the website, but if not, he will ask another server.

In case your ISP doesn't know the address, the request will be forwarded to the **root name server**.

Root name server – CERN Meyrin



Foto: Elias Wachmann

Now finally . . .

The root name server will send the IP address back to your ISP, which will send it back to your router, which will send it back to your computer.

Your computer now knows the address of the www.tugraz.at server and can send the actual request to load www.tugraz.at/institute/itpcp/

The server will send the requested website back to your computer, which will display it in your browser.

Ciphers

Encryption / Decryption

Encryption and decryption of data is performed to secure ones data from others.

It has to be fast to perform if the involved parties know the key for the encryption, but slow to crack if you don't hold the key.

Caesar Cipher

This is one of the oldest ciphers known to us. It's earliest known use was by Julius Caesar, hence the name.

The cipher employs a shift mechanism:

To encrypt: The letters of the alphabet are shifted by a given amount ($= z$) To decrypt: The letters are simply shifted backwards by the same amount

Example ($z = 3$):

Plaintext:	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>...</i>
Ciphertext:	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>	<i>k</i>	<i>l</i>	<i>...</i>

Caesar Example hints

A few hints for the Caesar example:

- There are three texts which were encrypted, all using Caesar Cipher
- Start with the first Text (not the whole file at once)
- Make sure to use the same alphabet as given in the file!!!
- Hints for the subsequent examples may be included :)